# An Application for Learning Chinese Language as Second Language: A Proposed Design

Nurul Harmiza Hipani[1] Saipunidzam Mahamad[2]

[1]Department of Computer and Information Sciences, Universiti Teknologi PETRONAS

Bandar Seri Iskandar, 31750 Tronoh, Perak MALAYSIA

armiza91@gmail.com[1], saipunidzam_mahamad@petronas.com.my[2]

**ABSTRACT.** *With the booming of mobile technologies and Internet connection nowadays, this paper presents a framework for learning Chinese language as a second language for primary school children aged 7 years old through a mobile application. In Malaysia, mobile learning technology still not widely used in the education system but in developing countries such as US had emerged into mobile learning technology in school. Plus, the non-Chinese students who take Chinese language as their second language in school have lower performance compared to the others. Hence, the main objective of this project is to encourage and support the process of learning Chinese language among young children who use Chinese as their second language at their own pace anytime anywhere by develop mobile learning application. It is designed in such a way of additional learning tool that young children can use and explore the modules in the application. From the user testing survey, most of the students would like to use mobile learning application in learning Chinese language as additional learning tool because the children find out it is interesting to learn new language via mobile. The application will include learning, exercise and also performance evaluation of the learners in interactive way. Thus, they will expose to the mobile technology learning at the early age with interesting play and learn application that will give excitement in learning the language. Indirectly, this mobile learning application also helps to boost the performance of the learners in learning a new language.*

**KEYWORDS --** Chinese language, mobile application, primary school children.

## INTRODUCTION

In the era of globalization, technology replaced many traditional aspects of our lives. It changed the methods of communication, altered the phenomenon of mobility and had a great impact on improving our lives in this emerging world. The advancement of mobile technologies emerge the dimensions of e-learning to m-learning. In principle, m-learning is a subset of e-learning which deals with learning through a mobile phone. There have been some researchers showed that using mobile devices in the course can improve learners' performance and arouse their motivation. The major advantages associated with mobile phones are the affordability, portability and interactivity.

This paper details one such application that deals with m-learning in learning a second language. Language is very important for a nation and every nation have their own formal language. Generally, a Malaysian is able to speak in at least two languages which are Malay and English. In learning a language, there are two main aspects which is memorizing and practicing in order for the learners to understand a vocabulary and its usage in the right context. The best age to learn a new language is young children because they are so much skilled at picking up a second language than adults [1]. It also does not hinder any abilities in the child's native language and they understand that language is something to explore, to play around with and to enjoy. Thus, starting early offers the widest possible set of benefits and opportunities.

In Malaysia, Minister of Education had introduced elective subjects such as *Bahasa Tamil Kebangsaan, Bahasa Cina Kebangsaan, Bahasa Arab, Bahasa Iban* and *Bahasa Kadazandusun* for primary school students. The objective for introduction of elective subjects in primary school is to attract the interest of students with different races to learn different language in early ages. Thus, the concept of 1Malaysia (1M) can be practiced and implemented in schools. For this paper, *Bahasa Cina Kebangsaan* (BCK) is chosen as a resource to develop the application for learning a second language. Based on *Pelan Induk Pembangunan Pendidikan 2006-2010, Bahasa Cina Kebangsaan* (BCK) is introduced in primary schools since 2005 which occupied approximately 14,000 students in 153 primary schools [2].

Therefore, this paper proposes an interactive design of mobile application that could help the non-Chinese students in primary school of age 7 years old to learn Chinese language as a second language according to the current syllabus of primary school education. The aim is to encourage young children to learn on daily usage of the language which are corresponded to the curriculum such as recognizing Chinese alphabets, numbers and objects. With the provided modules, learners will be able to measure their performance in mastering the language. However, this additional learning tool is not meant to replace the traditional way of teaching and learning in class but to create an excitement among the learners that promising more fun learning.

## RELATED WORK

The age factor in second language learning has long been, and continues to be a subject of much debate and controversy. Some researchers said that for those who learn a second language early in life will get higher levels of proficiency than those who begin as adults [3]. The ability of learning second language at early age is possible and would be a great advantage for a child even though children are thought using their native language since toddler. The learning experiences of a child determine which neural connections are developed and which no longer function. That means what is easy and natural for a child in learning a language [4]. An earlier start of learning a second language for young children enables productive links to be made between first and additional languages, which bring benefits for a child's language awareness and literacy [5]. Plus, they are likely to find it easier to acquire a good command of the sound system of the language, not only the pronunciation of individual sounds but also patterns of intonation. For children, they are always discovering new things and with a second language may encourage them on to a deeper and broader passion for learning in general.

Furthermore, numerous reports have proven that students who have studied a second language perform much better than monolingual peers on many assessments and standardized test in school. Results from the Scholastic Aptitude Test (SAT) show that students who had studied a foreign language for 4 or more years outscored other students on the verbal and math portions of the test [6]. Hence, it is an advantage for parents to send their kids to learn Chinese language as a second language to boost their kids' performance in other subjects. Learning at earlier stage also can maintain the interest of the children and they may easily catch up the learning with proper approach of teaching medium.

Based on the research survey in mobile learning, of over 900 learners who responded to an SMS survey, 91% agreed that mobile learning did help or may help them to learn; 93% believed that it did or sometimes did make learning more interesting; and 84% wanted to do more mobile learning in future [7]. From the survey, it can conclude that people are ready to accept mobile technologies due to the benefits could gain from it in teaching and learning. The main motivation behind the deployment of mobile phones in education is mobility. With mobile phones, students can access their lesson almost anywhere and at any time [7]. It will allow students to gather access and process information outside the classroom. Thus, they can encourage learning in real-world context, and help bridge school, afterschool and home environments [8].

Learning second language in more exciting and fun activities will help in improving children's skills on communication. In Malaysia, it is difficult to obtain mobile learning application that can attract children to learn foreign languages, especially Mandarin. Mandarin is the language that will be selected to be learned by majority of society. The number one reason is that Mandarin Chinese is the most widely-spoken language in the world which research stated that Mandarin Chinese is the mother tongue of over 873 million people [9]. Mandarin Chinese is also spoken in important and influential Chinese communities of Indonesia, Thailand, Malaysia, Singapore, Brunei and Philippines.

Therefore, learning Mandarin language in early education will bring benefits to the children itself in developing it careers in the future and also learning second language will differentiate them from the mono-lingual learner. Thus, in order to develop learning Mandarin language in early age, Ministry of Education (MOE) in Malaysia had introduced teaching and learning of Mandarin Language or also known as *Bahasa Cina Kebangsaan* (BCK) as elective subject in primary school and also secondary school.

## SYSTEM ARCHITECTURE

The conceptual architecture comprises two main stages. The first stage is the early analysis, which consists of problem identification and objective determination. Figure 1 shows the second stage in the application development which involves Rapid Application Development. The main objective of this proposed system is to develop an additional learning tool for non-Chinese students in the primary school to learn Chinese language which

is corresponded to the integrated curriculum. Thus, the content of the lessons are age-appropriate. Prototype will be tested to the real users to see the result and feedback from them.

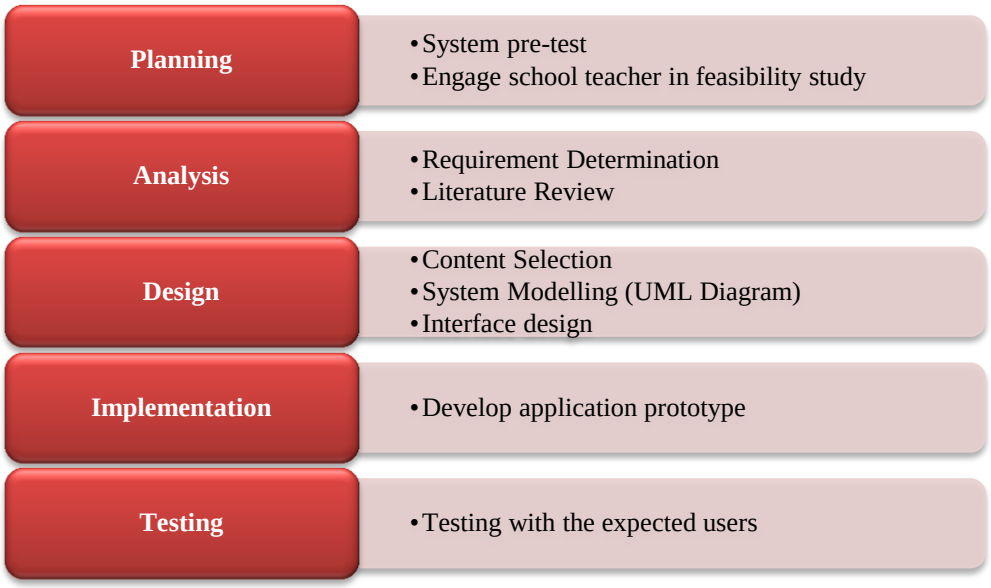



**FIGURE 1: Application Development**

Figure 2 shows the design of the system architecture for mobile learning application. From the figure, the users i.e. the teacher or students start the communication with the system by logging in and accessing the mobile learning application through mobile devices. The model also illustrates several levels of user authentications i.e. Student and teacher, which have different role of permission in accessing the contents and functionalities. Each of the users has to go through an authentication function before proceeding to the individual function such as learning, test and progress tracking.

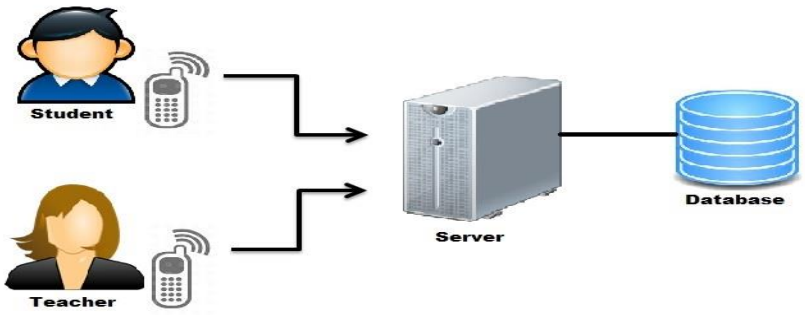



**FIGURE 2: The proposed system architecture**

Figure 3 shows the use case diagram for Learning Chinese Application System. Three different actors engage in this system which is the student, teacher and system administrator. It illustrates two main use cases which are login and choose function for the student. From Choose Function use case, the student can select learning, test or view performance from the system. There are several modules that can be chosen under learning function. Then, the student also can view his/her score after done with test from random modules. Meanwhile, teacher also has two main use cases which are login and also view students' performance. Lastly, the system administrator is the one that will manage the application in terms of improve the performance of the mobile learning application.

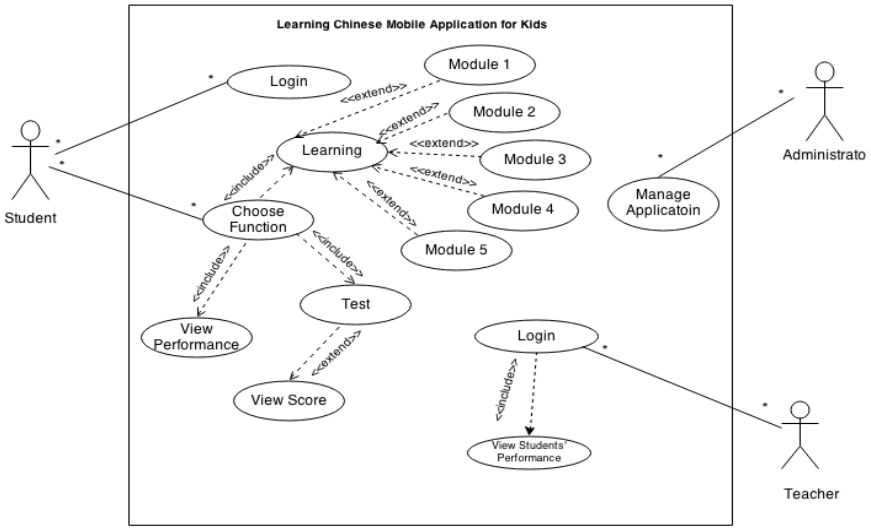



**FIGURE 3: Use Case Diagram**

**SYSTEM DESIGN**

By reviewing the common syllabus approved by the Ministry of Education under Curriculum Standard for Primary Schools (USSR) and discussion with teacher, there are five chosen modules for the development of the mobile learning application shown as below.

- **Module 1: Numeracy -** This module will teach student on recognize numeracy character in Chinese with pinyin and count from one to ten which is the basic level for the first year of primary level. The exercise will be provided at the end of the lessons where the students need to count the objects and select the correct answer.
- **Module 2: My Body -** This module will teach student about the part of human body such as eye, mouth, nose, hand and other parts in Chinese character with Pinyin. The exercise will be provided at the end of the lessons where the students need to match the part of the body with the correct Chinese character.
- **Module 3: My Family** - This module will provide the students about the members of family in Chinese character and pinyin. The exercise will be provided at the end of the lessons where the students need to match family members with the correct Chinese character.
- **Module 4 - My School** - In this module, the learning application will provide the students on things in class with different colors and sizes. Plus, it also include greeting between teacher and students.
- **Module 5 – Animals -** In this module, it will teach students on recognizing different type of animals in Chinese character with Pinyin. The exercise will be provided at the end of the lessons where the students need to match animals with the correct Chinese character.

Learning with question and answer approach will be used in developing the proposed system to make children feel more attractive and enjoy mobile learning. The designated mobile learning only involved simple questions as fill in the blank, matching the words, choose the correct answer. It is because the main goal of this application is to support learning Chinese language education among student in the early age and promote enjoyment of learning using mobile application. Therefore, children can learn by themselves without having active one-to-one learning process in the class. Plus, the usage of graphics and bright colors play an important role to make the application more interesting and students feel motivated.

There are three options that user might select in home screen which is choose lesson, test, and view score performance. The first choice will navigate user to the choose lesson screen whereby it contains list of module; in this application it involves five main modules. Then, users can select which module that they want to learn; Module 1 to Module 5. There will be a few exercises at the end of each module. User can choose the third option to see the overall performance. The overall process is shown in the Figure 1, 2 and 3 provided.

**MARI BELAJAR BCK APPLICATION**

The system interface was developed using Eclipse IDE and the contents in the application were collected from reliable source in the Internet and text book based on Primary School Standard Curriculum. Figure 4 shows the interface's screen shots of the application from an android device.

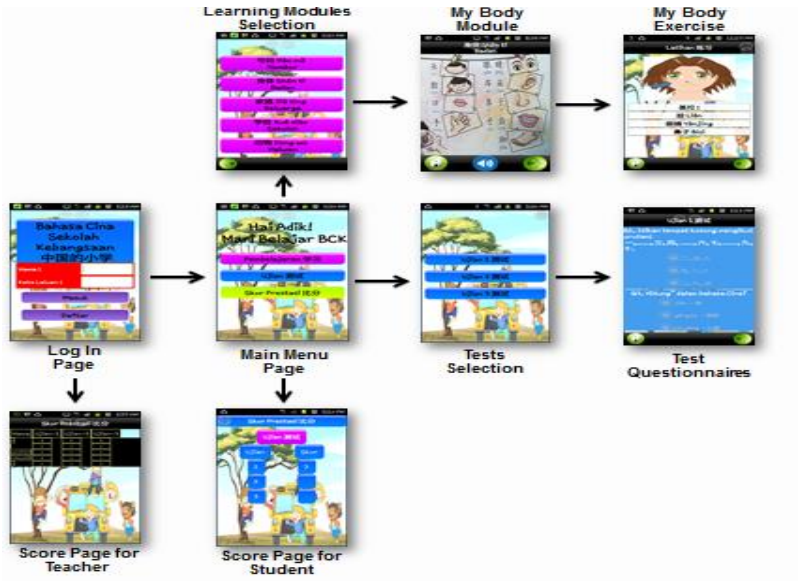



**FIGURE 4: Application's Interface Screenshots**

First of all, new user needs to create their account by register in the application. Meanwhile for existing user, he may directly log into his account and start exploring the application. The purpose is to track the progress for the test performance of each student registered. Once login success, the user will directly go to main menu of the application where he can select three options; learning, tests and score performance. In learning function, there will be five different modules which are numeracy, my body, my family, my school, and animals. At the end of each module, there will be several exercises for the users to answer. Then, the user also can select test function where there will be three different tests to be tested. The test will be in Multiple Choice Question (MCQ) where the user needs to click on the correct answer from several radio buttons. After done answering test and submit it, the user may check the results in score performance function. Same goes to teacher where the teacher can view the score performance for each of the students that had registered in the application.

**SYSTEM TESTING**

System testing was divided into two parts, which are developer testing and user testing. The project needs to be tested for conformance with the system requirements stated in the early stage. Below is the two parts involved:

**(a) Developer Testing**

The developer needs to test the application that deals with database where the user needs to register their account in the application and record their test performance. Log in function need to be synchronize with sign up details of the user account. Plus, the score performance obtained by the user for each test will be recorded and updated where the teacher can view it when he/she log into teacher account. Beside, testing also involves syntax, functionality and logical errors. No major problem found in this testing.

**(b) User Testing**

The system had been tested with school children when the system was almost completed. The system is tested by 10 students under the instructions of the teacher and the developer. Each student tested all the lessons and exercise with accompany by their friend. The non-Chinese students that taking BCK as their elective subject was selected as the novice users for this user testing. After the application prototype testing, the students need to fill in survey feedback form. The purpose of the survey is to collect the feedback from students regarding the usage of the application. There will be 4 questions in the feedback form which basically related to the interest of

students, user friendly application and also understandable modules. Figures 5 shows a group of student tested the application under developer supervision.



**FIGURE 5: Application System Testing**

## RESULTS AND DISCUSSION

The bar chart in Figure 6 shows the result of the user testing. The rating is from 1 to 5 which rating 1 signifies strongly disagree by the users and 5 signifies strongly agree by the users. It show that five out of ten respondents like the modules lesson in the application. They like to answer the questions provided at the end of each module. There are also familiar with the modules because they had learned it in class and the application is a revision for them to learn Chinese language anytime and anywhere they want. However, some of them dislike the modules in the application because it is limited and it is easy for them.

Moreover, six out of ten respondents seems want to try the application again if it is available in the future as additional learning tool to learn Chinese Language. This is because they found it is interesting to learn with new approach by using mobile device. However, there is a limitation for those who do not have smart phone and their parents not allowed them to use smart phone at their age. That is why some of them neither agree nor disagree to try the application again in the future.

Regarding the modules, the students agree that it can help them in learning Chinese Language based on the short lesson and exercises. But, two of the respondents feel that the lessons are not enough and they want to learn more with a lot of exercises. Lastly, the students not agree with the usability of the application. The student feel that the application is not easy to use because she had never experienced using smart phone before and then she find difficulties in using it for the first time.

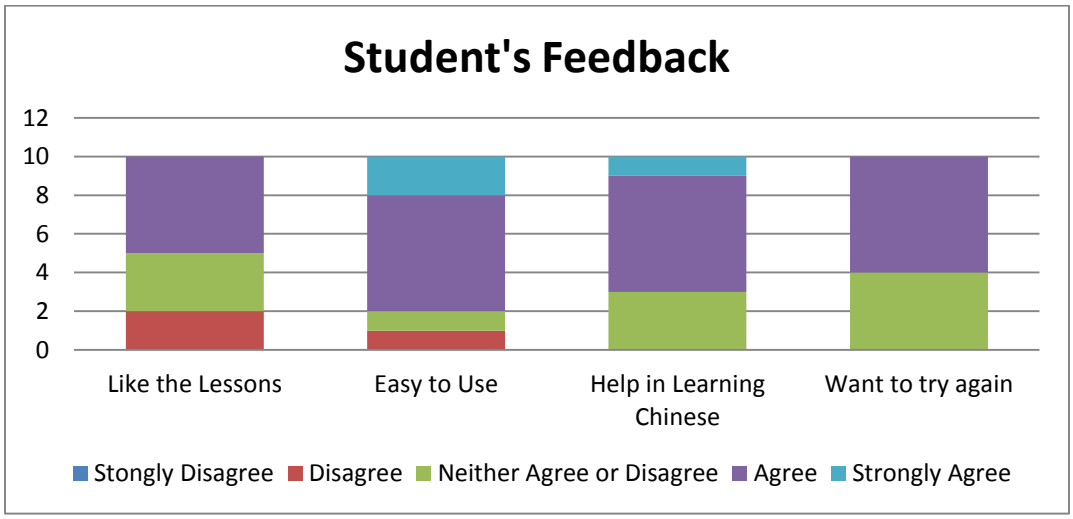



**FIGURE 6: Student's Feedback for Application System Testing**

Apart from the student's feedback, the teacher as the observer of the testing are also require to give some feedback regarding the application. Firstly, the teacher is required to explore the modules and ensure that all the modules are based on the learning syllabus of *Bahasa Cina Kebangsaan* Standard 1. Figure 7 shows that the teacher try and explore the application.

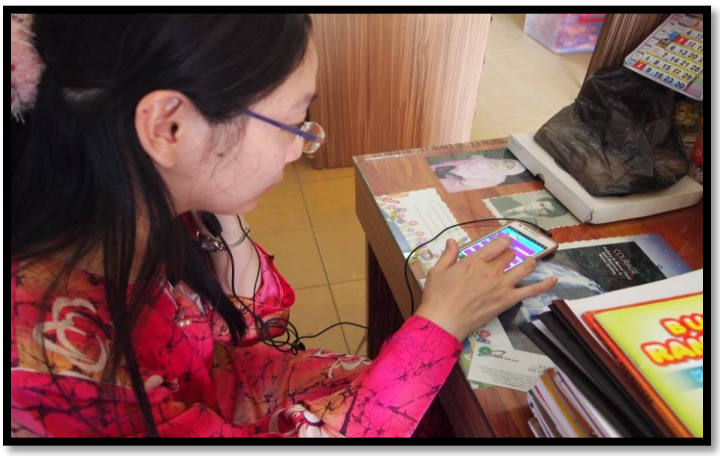


**FIGURE 7: Teacher Explore the Application**

From teacher's feedback, the application was able to attract student's interest based on the observation during user testing in the class. Furthermore, the application is ages appropriate due to the modules are made for Standard 1 student learning *Bahasa Cina Kebangsaan*. Plus, the modules are relevant to the content teaching in the classroom thus they can really apply it as an additional tool for learning BCK. The student's progress are also can be evaluated using this application since it has the function for the teacher to track the student score of each test they take. Lastly, the teacher agreed that the application is easy to use with the availability of narrator to guide the user on using the application. Therefore, the teacher believed that by using the application as additional learning tool, the student may maintain their interest in learning BCK as second language.

The development of *Mari Belajar BCK* Application could be enhanced in the future in order to add its capability to be preferred learning environment for primary schools. More research on Chinese learning modules and interactive exercise should be done. This will definitely enhance the modules to be more interactive and complex in terms of learning Chinese language. Besides that, uploading materials such as learning modules and exercise from computer also could be implemented to keep the students updated with different modules.

## CONCLUSION

In conclusion, the advancement in the field of Computer and Information Technology had broadened the scope of education in Malaysia. From traditional based learning environment, to electronic learning environment then upgrade to the new way of learning known as mobile learning that had been widely implemented in developed countries such as USA and UK. Ministry of Education in Malaysia should really pay a lot of attention to the mobile learning approach because it is getting popular nowadays as the technology advances day by day. With mobile learning, learning session can be done not only within class hours, but anytime and anywhere. Moreover, with this new approach of learning, learners may increase their interest of learning new language in the effective teaching environment by using mobile application as additional learning tools.

The preliminary study had been conducted in SK St. Bernadette's Convent, Batu Gajah to gain better understanding regarding the problem exist, the learning approach in learning BCK as second language for non-Chinese students and also modules to be included in the application. After three months developing the application, the prototype application is tested with 10 students in standard 1 taking BCK as their elective subject at the same school. The results of the testing showed the excitement of the student towards the application which is a good sign for the implementation as additional learning tool in the real learning environment in class in the future. Overall, this application may increase the interest of young children to learn BCK as their second language.

**ACKNOWLEDGEMENTS**

The author would like to acknowledge the support of this work to her FYP supervisor, Mr Saipunidzam Mahamad for his supervision in completing her Final Year Project. Besides, the acknowledgement also goes to Miss Wong Sook Fun, teacher who is teaching *Bahasa Cina Kebangsaan* in SK St. Bernadette's Convent for her willingness in helping the author by giving valuable information and suggestion.

**REFERENCES**

[1] Saipunidzam Mahamad, Fatimah Annor Ahmad Rashid, Mohammad Noor Ibrahim and Rozana Kasbon. "Mobile English Learning System: A Conceptual Framework for Malaysian Primary School".

[2] "Rancangan Malaysia ke – 9, Pelan Induk Pembangunan Pendidikan",2006-2010.

[3] Dwiana, A. and Singh. D. (2011). "Computer Game Based Learning Approach for Mandarin Language", in *International Conference on Electrical Engineering and Informatic*s: 1991

[4] Abdul Aziz, N.A., "Children's Interaction with Tablet Applications: Gestures and Interface Design", *International Journal of Computer and Information Technology (2),* 2013

[5] "The Benefits of Second Language Study". Research Findings with Citations. NEA Research (2007)

[6] Liu, S. "Modern Technologies in Chinese Language Teaching and Learning", in *IST Conference* (2002)

[7] Aubusson, P., Schuck, S., and Burden, K.. "Mobile Learning for Teacher Professional Learning: Benefits, Obstacles and Issues". *ALT-J Research in Learning Technology*. Vol.17 (No.3) pp 233-247, 2009.

[8] Saipunidzam Mahamad, Mohammad Noor Ibrahim and Shakirah Mohd Taib. "M-Learning: A New Paradigm of Learning Mathematics in Malaysia" *International Journal of Computer Science & Information Technology* (IJCSIT), Vol.2 (No.4). pp. 76-78.

[9] Tseng, C.C., Lu, C.H. and Hsu, W.L. "A Mobile Environment for Chinese Language Learning", *Current Developments in Technology-Associated Education* (2006).